

MSE 300 Exam 2 Study guide

Topic 5: Crystallography in 1D & 2D

- Symmetry elements and operations in 1D and 2D
- Application of symmetry operations
- Proper vs improper operations
- Decoding line groups and plane groups, including Bravais lattices and symmetries
- Identifying the lattice points, unit cell, basis, point group and line/plane group in 1D/2D arrays

Topic 6: 3D Crystal symmetries, Point Groups and Space Groups

- Symmetry elements and operations in 3D
- Application and combination of symmetry operations
- Proper vs improper operations
- Point groups and how to decode them (crystal system, symmetries, their directions)
- Can distinguish centrosymmetric from non-centrosymmetric point/space groups
- Interpret point group planar projections
- Calculate general equivalent positions from a point group
- Derive point groups, crystal systems, Bravais lattices and (non)centrosymmetry from space groups
- Information encoded by a point group and/or space group and its general interpretation
- Calculate general equivalent positions for a given space group
- Order of a space group and multiplicity
- Interpretation of International Tables of Crystallography, including drawing the full unit cell with atomic positions

Topic 7: Crystallographic computations in 3D

- Compute the metric tensor of a Bravais Lattice
- Calculate the distance between any two points in Bravais lattices
- Calculate the angle of a bond or between any two directions in Bravais lattices